Difference Between Inverse Functions and Inverse Images

Not every function has an inverse function. In order for $f : X \to Y$ to have an inverse, f must be one-to-one and onto. However, for ANY function, the inverse image of ANY subset of the target is defined. Unfortunately, the notation for inverse function is part of the notation for inverse image, so that you have to determine from context which meaning is meant. The basic definition of inverse image is

Definition: Let $f : X \to Y$ be a function. Let $U \subset Y$. Then the *inverse image of* U, denoted $f^{-1}(U)$, is the set

$$f^{-1}(U) = \{x \in X \mid f(x) \in U\}.$$

Note that it is only the entire character-string "$f^{-1}(U)$" that is defined here; the sub-string "f^{-1}" has no independent meaning in this context (just as the letter d in *dog* has no independent meaning). Note also that the inverse image of a set is a SET, not an element (although it is possible for the inverse image of a set to contain only one element).

It's important to realize that you do not need a function to be 1-1 or onto for inverse images to make sense.

Examples Take $X = Y = \mathbf{R}$, and let $f(x) = \cos x$. Then

1. $f^{-1}(1)$ makes no sense, since "1" is not a subset of the target. (It is an *element* of the target.)
2. $f^{-1}(\{1\}) = \{$all integer multiples of $2\pi\}$.
3. $f^{-1}([0,1]) = \bigcup_{n\in\mathbf{Z}}[2n\pi - \pi/2, 2n\pi + \pi/2]$. (Recall that $\mathbf{Z}$ is the standard notation for the set of all integers, and that $[a, b]$ means the closed interval in $\mathbf{R}$ from a to b.)
4. $f^{-1}(\{2\}) = \emptyset$. (This is my word processor's notation for the empty set, not zero.)
5. $f^{-1}((0, 63]) = \bigcup_{n\in\mathbf{Z}}(2n\pi - \pi/2, 2n\pi + \pi/2)$.
6. $f^{-1}([-1, 1]) = \mathbf{R}$.

Despite what I said in the first example, it is common to "abuse terminology" or "abuse notation" and refer to the inverse image of a single element, when one what actually means is the object illustrated in the second example. Even with this abuse of terminology, the inverse image of an element is understood to be a set. If $f : X \to Y$ and $y \in Y$, then any element of the set $f^{-1}(y)$ (abuse of notation!—I should really write $f^{-1}(\{y\})$) is called *an* inverse image of y. For example, for the case of the cosine function above, 4π is an inverse image of 1.

If a function $f : X \to Y$ happens to have an inverse function, then it is true that $f^{-1}(U) = \{f^{-1}(y) \mid y \in Y\}$ (here the first use of "f^{-1}" is part of the notation for the inverse image of a set, while the second use refers to the function called "f inverse". If f has an inverse, and U consists of a single element of Y, then the notion of inverse image reproduces the notion of inverse function:

$$f^{-1}(\{y\}) = \{f^{-1}(y)\}, \quad \forall y \in Y.$$